



Ministero dell'Istruzione, dell'Università e della Ricerca

01AA90010

ADMISSION TEST FOR THE DEGREE COURSE IN MEDICINE AND SURGERY

Academic Year 2019/2020



Cambridge Assessment
Admissions Testing

1 At the Shawilde Theatre, tickets are priced as follows:

Front stalls £35
Centre stalls £28
Rear stalls £20

Every row in the theatre contains 30 seats.

Last night's performance was a sell-out and the ticket sales were £21 000. Exactly 60% of this income came from tickets for the centre stalls, and the rest was split equally between the front stalls and the rear stalls.

How many rows of seats make up the centre stalls at the Shawilde Theatre?

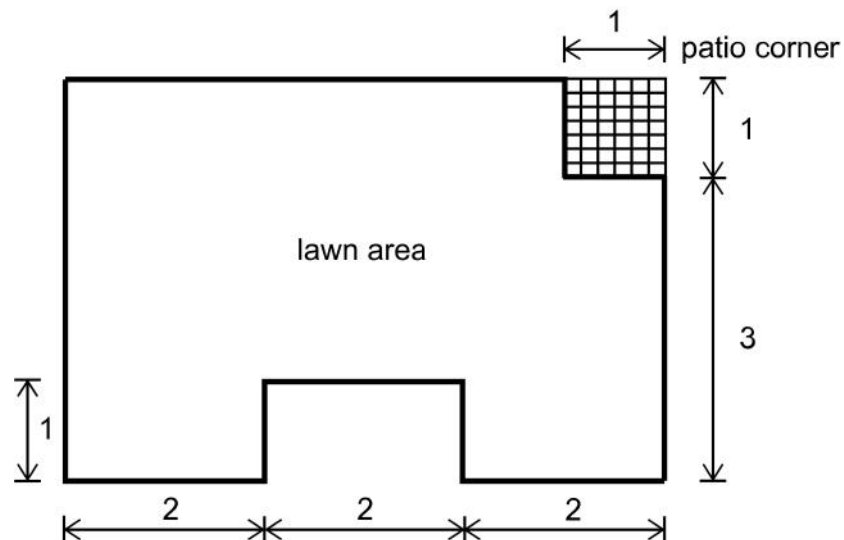
- A** 19
- B** 15
- C** 22
- D** 26
- E** 11

- 2** Increasingly, the quality of a teacher, and of his or her lessons, is to be judged by feedback from their pupils. This is dangerous, as pupils have a tendency to focus too much on whether or not they enjoyed the lesson, in other words, on how much 'fun' they found it. The purpose of education, however, is not to keep children entertained; it is not to make their lives more fun. Arguably, it is not even to make them happier, per se. The purpose of schools is to make their pupils better, more educated people. This is all that matters, and therefore it is whether the teacher succeeds at this – not at being a children's entertainer – that counts.

Which one of the following is a flaw in the above argument?

- A** By simply asserting that the purpose of education is not even to make people happier, the argument rests on a questionable contention.
- B** It doesn't specify precisely what it means by making someone a better, more educated person.
- C** The comparison with being a children's entertainer is an unfair exaggeration of what a teacher tries to do by making lessons more fun.
- D** It fails to consider that enjoying lessons could result in pupils ending up better educated.
- E** It patronises young people by thinking that their judgement will be based solely on how much fun they found the lesson.

- 3 The plan of a domestic lawn with dimensions in metres is shown below.



The lawn area is to be reseeded. Seed must be applied at the rate of 50 grams per square metre.

Appropriate grass seed is available only in the following quantities:

grass seed packet prices	
1.6 kg	£20.00
1.2 kg	£16.50
1.0 kg	£14.00
500 g	£8.00
200 g	£3.00
100 g	£2.00

What is the least amount of money that needs to be spent to buy sufficient seed to reseed the lawn?

- A £16.50
- B £13.50
- C £14.00
- D £16.00
- E £22.00

- 4** Once again it has become fashionable for householders to replace their carpets with wooden floors. Sales of laminate and solid floors, such as oak, have seen a massive increase in the last ten years. This trend seems more in tune with our eco-friendly aspirations: carpets are often plastic-based and use vast quantities of underlying material, or underlay, made from petrochemicals. For this reason, it seems unlikely that wooden floors will go out of fashion in the near future.

Which one of the following, if true, most weakens the above argument?

- A** Carpets can harbour dust mites and allergens.
- B** Not all wooden floors need a layer of underlay.
- C** Most types of laminate and wood floors use petrochemicals in their manufacture.
- D** Wooden floors may look longer-lasting but can be more easily damaged than carpets.
- E** Sales of carpets made of synthetic materials have decreased in the last ten years while woollen carpets have seen an increase.

- 5 Five girls competed in five events at their sports day. The following scoring system was used for each event:

1 st place	6 points
2 nd place	3 points
3 rd place	1 point
4 th /5 th place	0 points

Before the events started, each girl nominated the event in which they wished to play their joker card – this doubled the number of points gained in that event. The table below shows the results of each event and the athletes' final points scores:

<i>competitor</i>	<i>800 m run</i>	<i>100 m hurdles</i>	<i>high jump</i>	<i>long jump</i>	<i>shot put</i>	<i>total points scored</i>
<i>Tia</i>	4 th	1 st	5 th	3 rd	3 rd	14
<i>Jessica</i>	1 st	3 rd	1 st	5 th	4 th	19
<i>Carolina</i>	3 rd	4 th	2 nd	1 st	2 nd	14
<i>Kelly</i>	2 nd	2 nd	4 th	2 nd	5 th	12
<i>Helena</i>	5 th	5 th	3 rd	4 th	1 st	13

How many of the girls won the event in which they had elected to play their joker?

- A 4
- B 2
- C 3
- D 1
- E 5

- 6** A study involving a brain-training exercise was carried out on more than a thousand adults aged 65 and over, some of whom later developed dementia. Results showed that the benefits of the five-week mental agility course undertaken by some of the adults lasted for at least five years. This led to an improvement in everyday activities such as money management and the ability to do housework. If those with trained brains developed dementia, they did so later than those in the control group. The results also showed that, for those people in the study who developed dementia, after the diagnosis their mental decline occurred faster than for those who had not undertaken the training.

Which one of the following can be drawn as a conclusion from the above passage?

- A** People do a decreasing amount of housework as they grow older.
- B** It is preferable to have swift mental decline once dementia develops.
- C** Older people do not perform mentally challenging tasks unless forced to do so.
- D** Keeping the mind active delays the onset of dementia.
- E** All over-65s who undertake brain training live for at least five years afterwards.

- 7** According to the current mainstream scientific view, Near Death Experiences (NDEs) are explicable in purely physiological terms. Specifically, they are caused by cerebral anoxia (oxygen deficiency in brain tissue), which occurs in a dying brain. On the other hand, recent research on hundreds of successfully resuscitated cardiac patients found that only twenty per cent reported NDEs. If NDEs had purely medical causes then most of the patients should have experienced them, since they had all been clinically dead and experienced cerebral anoxia. NDEs therefore do not have purely physiological causes.

Which one of the following best expresses the main conclusion of the above passage?

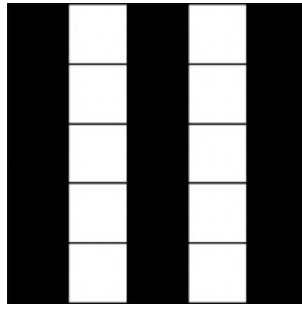
- A** Not all successfully resuscitated cardiac patients have NDEs.
- B** Not all clinically dead patients have NDEs.
- C** NDEs are caused by oxygen deficiency in the brain.
- D** NDEs are not necessarily caused by physical events alone.
- E** NDEs are a physical property of the human brain.

- 8 A student packs books into a box which is 20 cm high, 30 cm wide, and 40 cm long. She ties string around the box in the pattern shown. A further 20 cm of string is needed for a knot.



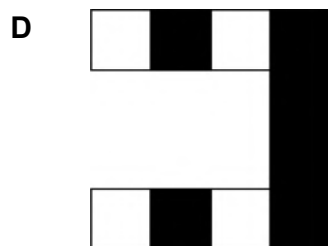
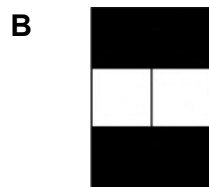
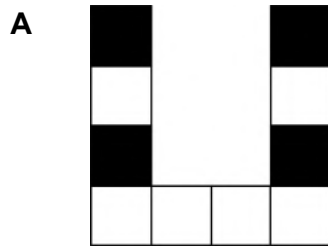
What length of string is needed to tie up her box?

- A 2.6 m
- B 5.4 m
- C 2.9 m
- D 3.8 m
- E 2.0 m



Four of the following five pieces can be fitted together to make the above pattern.

Which one of the pieces is NOT needed?



- 10** Researchers have tried to establish reasons for a recent drop in the catch of marlin near the Madeira coast. This drop cannot be explained by environmental conditions; there was no significant change in the temperature or composition of the region's water. Nor could one blame the human factor, because the size of the fishing fleet, catch quotas and pollution levels have been strictly observed. The environmentalists then analysed the records of the fishing catch of marlin over the last six centuries. This research identified that regular five-year periods of significant decrease in the catch alternated with fifteen-year periods of gradual growth of the catch. In view of this, the study concluded that the current drop in the marlin catch is part of a long-term cycle related to fish migration and food-chain balance.

Which one of the following is an underlying assumption of the above argument?

- A** There is no correlation between the fishing catch today and the deep-sea stock of marlin in the sixteenth century.
- B** The fishing catch of marlin around Madeira does not depend on the condition of the water around the island.
- C** Climate change did not affect the temperature of ocean water off the Madeira coast.
- D** The change in fishing technology has affected the change in the fishing catch.
- E** The records provide accurate information about the catch of marlin off the Madeira coast.

- 11** Which one of the following writers is the author of the novel *To Kill a Mockingbird*?

- A** Iris Murdoch
- B** Harper Lee
- C** Elena Ferrante
- D** Georges Simenon
- E** Ernest Hemingway

- 12** Which one of the following is NOT a stated purpose of the United Nations?
- A** To achieve international co-operation in solving international problems
 - B** To encourage peaceful trade between nations
 - C** To develop friendly relations among nations
 - D** To maintain international peace and security
 - E** To be a centre for harmonizing the actions of nations
- 13** Which one of the following composed the opera *Madama Butterfly*?
- A** Giacomo Puccini
 - B** Richard Wagner
 - C** Georges Bizet
 - D** Gioachino Rossini
 - E** Giuseppe Verdi
- 14** Which one of the following countries did NOT adopt the coins and banknotes of the Euro as its currency on 1 January 2002?
- A** Finland
 - B** Austria
 - C** Portugal
 - D** Luxembourg
 - E** Sweden

15 Which one of the following plays by William Shakespeare is NOT set in Italy?

- A** A Midsummer Night's Dream
- B** Romeo and Juliet
- C** Much Ado About Nothing
- D** Othello
- E** The Taming of the Shrew

16 Which one of these events in world history happened most recently?

- A** The building of the Taj Mahal
- B** The crowning of Charlemagne
- C** The October Revolution in the Russian Empire
- D** The Taiping Rebellion in China
- E** The fall of the Western Roman Empire

17 Who was the first woman to be awarded a Nobel Prize?

- A** Selma Lagerlöf
- B** Grazia Deledda
- C** Mother Teresa
- D** Marie Curie
- E** Doris Lessing

- 18** The 'prisoner's dilemma' is a common example analysed in the field of
- A** Quantum theory
 - B** Chaos theory
 - C** Asymptotic theory
 - D** Network theory
 - E** Game theory
- 19** Which one of the following religions is NOT considered to have been revealed to or founded by an individual?
- A** Confucianism
 - B** Hinduism
 - C** Islam
 - D** Buddhism
 - E** Christianity
- 20** The Constitution of the Italian Republic was enacted on 22 December 1947 by
- A** The Government
 - B** The Ministry of Defence
 - C** The Constituent Assembly
 - D** The Ministry of Justice
 - E** The Supreme Court

21 Which one of the following literary works does NOT originate in the corresponding country?

- A** The Divine Comedy – Italy
- B** Oedipus Rex – Greece
- C** The Poems of Rumi – China
- D** Don Quixote – Spain
- E** The Tale of Genji – Japan

22 The governance of modern states is often based on the separation of which three powers?

- A** Legislative – elective – executive
- B** Regulatory – executive – judicial
- C** Electoral – regulatory – jurisprudential
- D** Legislative – elective – jurisprudential
- E** Legislative – executive – judicial

23 An electron microscope (EM) was used to view a cell.

The maximum dimension of the cell was observed at a magnification $\times 30\,000$. The image on the EM screen showed a maximum length of 30 mm.

Which of the following mature healthy cells was being magnified?

- A** one coccus bacterium in a Staphylococcus cluster
- B** one palisade cell from a wheat plant
- C** one mature human red blood cell
- D** one sensory neuron from a human
- E** one lymphocyte from a human

24 Which row is correct for the three given features of typical human B lymphocytes?

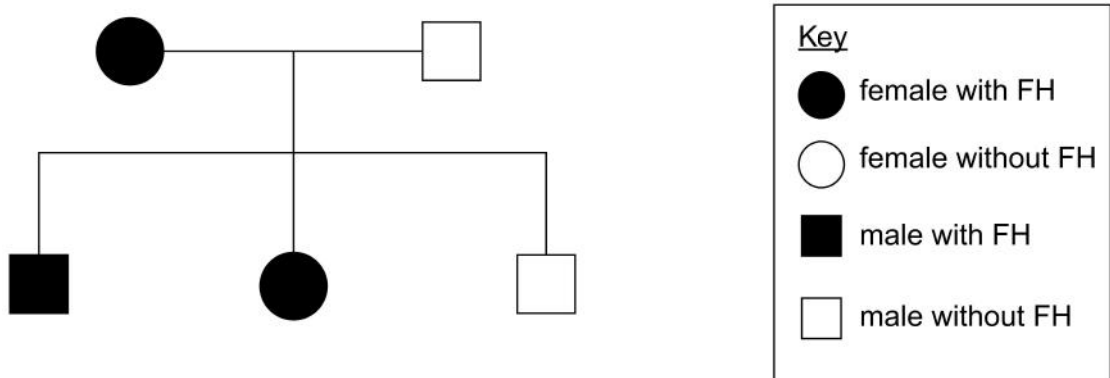
Key
✓ correct
x incorrect

	<i>formed from bone marrow stem cells</i>	<i>processed in the thymus</i>	<i>contain genes coding for antibodies</i>
row 1	✓	✓	✓
row 2	✓	✓	x
row 3	✓	x	✓
row 4	x	x	✓
row 5	x	✓	x

- A** row 3
- B** row 5
- C** row 4
- D** row 2
- E** row 1

- 25** Familial hypercholesterolaemia (FH) is an inherited disease caused by a dominant allele. People with FH have high concentrations of cholesterol in the blood, which can lead to an increased risk of coronary heart disease.

The family tree shows the inheritance of this disease in one family.



Which row shows the number of individuals in the family tree with each genotype?

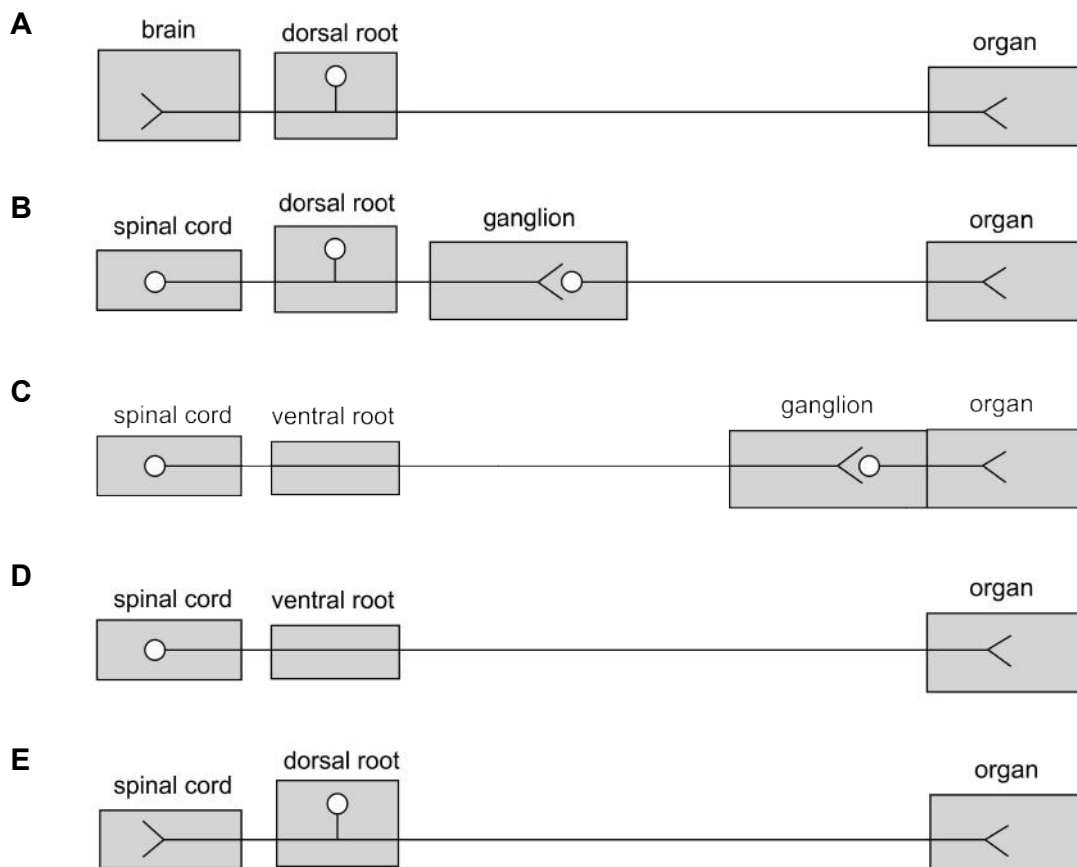
	<i>genotype</i>		
	<i>homozygous dominant</i>	<i>homozygous recessive</i>	<i>heterozygous</i>
row 1	3	2	0
row 2	2	3	0
row 3	0	2	3
row 4	0	3	2
row 5	3	0	2

- A** row 2
- B** row 1
- C** row 3
- D** row 4
- E** row 5

- 26** Which one of the following occurs during anaphase of mitosis in a healthy human liver cell?
- A** Chromatin condenses so that the chromosomes are visible under the microscope.
 - B** Centrioles migrate to opposite poles of the cell and help to assemble the spindle fibres.
 - C** DNA replicates so that each chromosome is made up of two sister chromatids.
 - D** Sister chromatids are pulled by the spindle fibres towards opposite poles of the cell.
 - E** Chromosomes move towards the equator of the cell and attach to the spindle fibres.

- 27** The diagrams show some pathways between the central nervous system and effector organs in a healthy human.

Which pathway is a possible route for impulses passing along a parasympathetic neuron?



28 Which row shows a correct component of each of the three structures listed?

	<i>viral envelope</i>	<i>capsid</i>	<i>genetic material</i>
row 1	glycogen	glycosidic bonds	adenine
row 2	glycosidic bonds	glycogen	ester bonds
row 3	phosphodiester bonds	hydrogen bonds	amino acids
row 4	fatty acids	amino acids	phosphodiester bonds
row 5	phosphate	phosphodiester bonds	cytosine

- A** row 5
- B** row 4
- C** row 3
- D** row 1
- E** row 2

29 An animal has a coat that has some patches of white hair and some patches of black hair. Cells that make up a hair follicle form a single hair, which is either white or black.

Assume that there are no mutations following formation of the zygote.

A student wrote the following statements:

- 1** The cells in all the hair follicles contain the same genes.
- 2** The alleles in a hair follicle forming a white hair are different from the alleles in the hair follicle forming a black hair.
- 3** Each white hair follicle cell contains two recessive alleles.

Which statement(s) is/are correct?

- A** 2 and 3 only
- B** 2 only
- C** 1, 2 and 3
- D** 1 only
- E** 1 and 2 only

30 Which of the following is/are tissues?

- 1** cartilage
- 2** skin
- 3** endothelium

- A** 2 and 3 only
- B** 1 and 2 only
- C** 1 only
- D** 1, 2 and 3
- E** 1 and 3 only

31 The same section of an allele coding for five amino acids from two different individuals, P and Q, is shown.

P	ATTCCGGGATTCCCT
Q	ATTCCGGATTGCACT

Which of the following types of mutation could explain the differences between P and Q?

- 1** addition
- 2** deletion
- 3** substitution

- A** 1, 2 and 3
- B** none of them
- C** 1 and 3 only
- D** 1 only
- E** 2 and 3 only

- 32 Beetroot is a root vegetable with cells that contain a red pigment. Normally the pigment cannot pass out of the cells because it cannot diffuse through their cell surface membranes. An investigation was carried out into the effect of various chemicals on the permeability of the cell surface membranes of beetroot cells.

1 cm³ samples were cut from the beetroot and washed in running water for 20 minutes to remove any pigment that was released from the damaged cells.

Five experiments were carried out. In each experiment, one cube of beetroot was placed in liquid P and one cube placed in liquid Q, and the results were observed.

Which row shows the correct results and conclusions?

Key
✓ does happen
x does not happen

		<i>result</i>	<i>conclusion</i>		<i>result</i>	<i>conclusion</i>
	<i>liquid P</i>	<i>liquid P turns red</i>	<i>phospholipids dissolve</i>	<i>liquid Q</i>	<i>liquid Q turns red</i>	<i>protein denatures</i>
row 1	dilute HCl	x	x	water at 20 °C	✓	✓
row 2	dilute HCl	✓	✓	ethanol	✓	✓
row 3	ethanol	✓	✓	dilute HCl	✓	✓
row 4	ethanol	x	x	water at 20 °C	✓	✓
row 5	water at 20 °C	✓	✓	dilute HCl	x	x

- A** row 2
- B** row 5
- C** row 1
- D** row 4
- E** row 3

33 Which of the following rows are correct about these naturally occurring biological molecules?

	<i>biological molecule</i>	<i>presence of carboxylic acid groups</i>	<i>presence of 2 or more hydroxyl groups</i>
row 1	ALL amino acids	yes	no
row 2	glucose	no	yes
row 3	glycerol	no	no
row 4	ALL fatty acids	yes	no

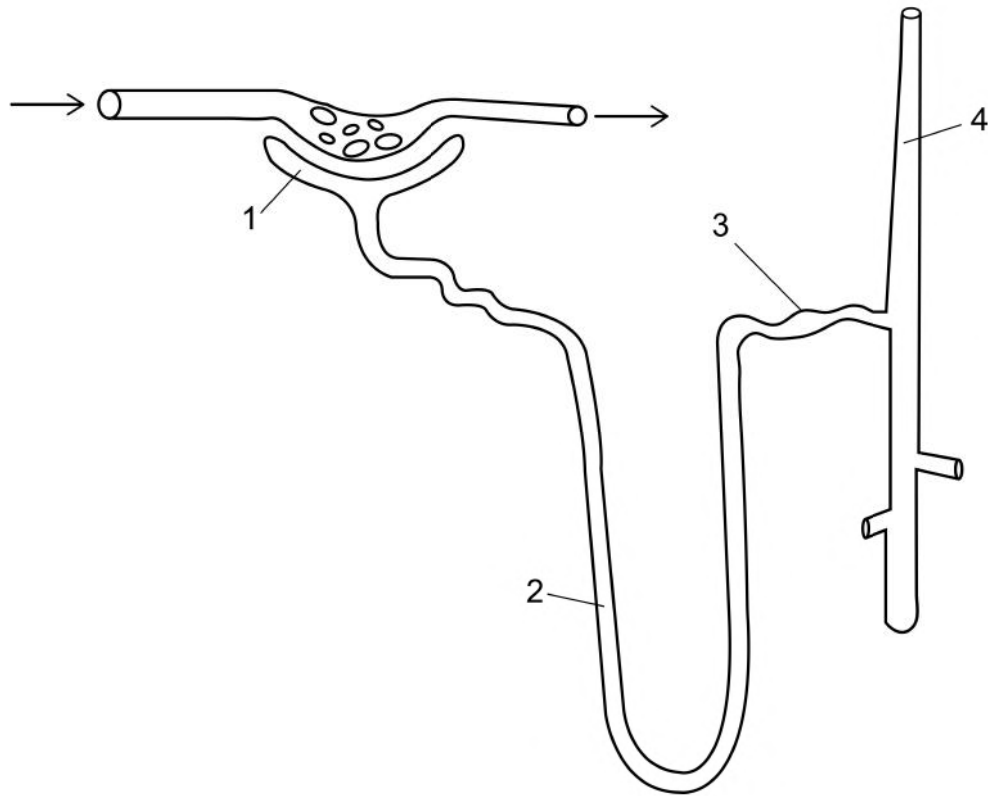
- A** 1, 2 and 4 only
B 1, 2, 3 and 4
C 2, 3 and 4 only
D 1, 3 and 4 only
E 1, 2 and 3 only

34 Which of the following regions within a sarcomere remain UNCHANGED in length when a healthy human muscle cell contracts?

- 1** A-band
2 I-band
3 H-zone / band

- A** 2 only
B 3 only
C 2 and 3 only
D 1 only
E 1 and 3 only

The diagram shows a nephron with collecting duct from a healthy human.



Which of the regions (1-4) contain a liquid with no urea?

- A** 1 only
- B** 1 and 2 only
- C** 2 and 3 only
- D** 2, 3 and 4 only
- E** none of them

36 Which of the following statements about the metabolism of plants is/are correct?

- 1** CO₂ is produced during the day.
- 2** CO₂ is produced at night.
- 3** The CO₂ produced can be used by the plant.

- A** 1 and 2 only
- B** 1 only
- C** 2 and 3 only
- D** 2 only
- E** 1, 2 and 3

37 Which statement about an enzyme involved in genetic modification is correct?

- A** A ligase causes H-bonds to form between sticky ends only.
- B** A restriction enzyme causes only H-bonds to be broken.
- C** A restriction enzyme causes phosphodiester bonds to be broken.
- D** A ligase causes phosphodiester bonds to be broken.
- E** A restriction enzyme causes H-bonds to form between sticky ends.

38 If a cell divides by mitosis ten times, what is the number of cells at the end of the process?

[Assume that all cells remain alive.]

- A** 64
- B** 11
- C** 10
- D** 100
- E** 1024

- 39** Which statement is correct in healthy humans?
- A** Increasing the partial pressure of oxygen makes it more likely that haemoglobin will release its oxygen.
 - B** Blood leaving active muscles that are respiring aerobically will contain hydrogencarbonate ions and raised levels of lactic acid.
 - C** In active tissues where the carbon dioxide concentration is high, haemoglobin has a low affinity for oxygen.
 - D** The majority of carbon dioxide transported in the blood is in the form of carbaminohaemoglobin.
 - E** At a low partial pressure of oxygen, myoglobin is less saturated with oxygen than haemoglobin.

- 40** Which of the following processes involve ions?
- 1** contraction of a muscle
 - 2** transmission of a nerve impulse across a synapse
 - 3** transfer of light energy into chemical energy in photosynthesis
 - 4** oxidative phosphorylation in a mitochondrion
- A** 2 and 3 only
 - B** 1 and 2 only
 - C** 3 and 4 only
 - D** 1, 2 and 3 only
 - E** 1, 2, 3 and 4

- 41** A student observed that when powdered sugar at -1°C was added to ice at -1°C all of the ice melted.

Which of the following statements explain this observation?

- 1** Sugar has given the water molecules energy.
- 2** The melting point of the mixture is higher.
- 3** The melting point of the mixture is lower.

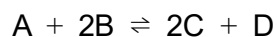
- A** 1 only
B 1 and 3 only
C 3 only
D 2 only
E 1 and 2 only

- 42** Which of the following equations represent(s) a redox reaction?

- 1** $4\text{LiH} + \text{AlCl}_3 \rightarrow 3\text{LiCl} + \text{LiAlH}_4$
- 2** $\text{N}_2\text{O}_3 + 3\text{H}_2\text{O} \rightarrow 2\text{H}_3\text{O}^+ + 2\text{NO}_2^-$
- 3** $\text{NH}_4\text{NO}_3 \rightarrow 2\text{H}_2\text{O} + \text{N}_2\text{O}$

- A** 3 only
B 1, 2 and 3
C 1 and 3 only
D 1 and 2 only
E 2 only

- 43** Consider the following reversible reaction at temperature T .



At equilibrium, there are 0.5 moles of A, 0.2 moles of B, 0.5 moles of C and 0.8 moles of D, all in a vessel of volume V .

What is the value of the equilibrium constant, K_c , at this temperature?

- A** 0.1
- B** 2
- C** 10
- D** 4
- E** 0.25

- 44** A compound consists of carbon, hydrogen and oxygen only.

8 g of this compound contains 3 g of carbon and 1 g of hydrogen.

Which one of the following could be the molecular formula of the compound?

[A_r values: C = 12; H = 1; O = 16]

- A** CH_2O
- B** C_3HO_4
- C** $\text{C}_2\text{H}_8\text{O}$
- D** $\text{C}_2\text{H}_4\text{O}$
- E** CH_4O

45 Why does hydrogen iodide (HI) have a higher boiling point than hydrogen bromide (HBr)?

[Atomic numbers: bromine = 35; iodine = 53]

- A** The iodide ions in HI are larger than the bromide ions in HBr.
- B** The H – I covalent bond is stronger than the H – Br covalent bond.
- C** There are hydrogen bonds between HI molecules but not between HBr molecules.
- D** The permanent dipole-permanent dipole forces between HI molecules are stronger than those between HBr molecules.
- E** The induced dipole-induced dipole (dispersion) forces between HI molecules are stronger than those between HBr molecules.

46 Element Z reacts with water at room temperature and hydrogen gas is released.

The oxide of element Z is a solid at room temperature and does not conduct electricity. The oxide reacts with acids and has the formula ZO.

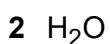
To which group/part of the Periodic Table does element Z belong?

- A** Group VII
- B** Transition metals
- C** Group I
- D** Group III
- E** Group II

47 Which statement about a positive ion explains why it is positively charged?

- A** The ion has the same number of protons and neutrons.
- B** The ion has more electrons than protons.
- C** The ion has more protons than electrons.
- D** The ion has more electrons than neutrons.
- E** The ion has more protons than neutrons.

48 Which of the following molecules have an overall permanent dipole moment?



[C is in Group IV, N is in Group V, O is in Group VI]

A none of them

B 1 and 2 only

C 1, 2 and 3

D 2 and 3 only

E 1 and 3 only

49 A few drops of universal indicator solution were added to exactly 50 mL of sodium hydroxide solution, NaOH, of concentration 1 mol / L. Exactly 50 mL of hydrochloric acid, HCl, of concentration 1 mol / L, was added drop by drop to the sodium hydroxide solution containing the universal indicator solution.

What colour changes would be observed as the hydrochloric acid was added?

A red → orange → yellow → green

B blue → red

C red → orange → yellow → green → blue → violet

D violet → blue → green

E violet → blue → green → yellow → orange → red

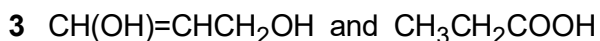
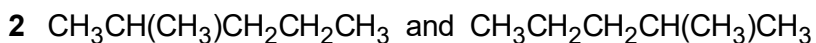
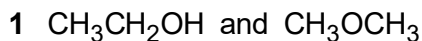
50 Hydrogen has a radioactive isotope called tritium. Tritium atoms have two neutrons.

The most abundant naturally occurring isotope of hydrogen has a relative atomic mass of 1.0.

What is the value for the relative atomic mass of a sample of hydrogen gas that has an equal amount of these two isotopes of hydrogen?

- A** 3.0
- B** 2.0
- C** 4.0
- D** 1.5
- E** 1.0

51 Which of the following pairs of structural formulae represent structural isomers?



- A** 2 only
- B** 1 only
- C** 1, 2 and 3
- D** 1 and 3 only
- E** 2 and 3 only

- 52** What is the minimum mass of sulfur dioxide needed to make 500 mL of a solution of concentration 3.0 mol / L?

[A_r values: O = 16; S = 32]

- A** 32 g
- B** 48 g
- C** 192 g
- D** 384 g
- E** 96 g

- 53** A car, which is initially stationary, accelerates for 5.0 seconds at 4.0 m/s^2 along a straight road. It then continues in the same direction for 20 seconds at a constant speed.

What is the maximum speed of the car, and what is the distance travelled by the car in the final 20 seconds of its motion?

- A** maximum speed = 24 m/s ; distance = 480 m
- B** maximum speed = 20 m/s ; distance = 400 m
- C** maximum speed = 9.0 m/s ; distance = 200 m
- D** maximum speed = 9.0 m/s ; distance = 180 m
- E** maximum speed = 20 m/s ; distance = 80 m

- 54** The mean mass of three babies is 2.1 kg . The range of their masses is 0.7 kg . The lightest baby has a mass of 1.8 kg .

What is the median mass of the three babies?

- A** 2.1 kg
- B** 2.3 kg
- C** 2.15 kg
- D** 2.0 kg
- E** 1.95 kg

- 55** A stone of density 5.20 g / cm^3 and volume 200 cm^3 is completely submerged in a liquid of density 1.20 g / cm^3 .

What is the magnitude of the upthrust acting on the stone?

[gravitational field strength = 10.0 N / kg]

- A** 6.40 N
- B** 2.00 N
- C** 10.4 N
- D** 8.00 N
- E** 2.40 N

- 56** A fixed mass of an ideal gas is compressed at constant temperature. The pressure is recorded continuously as the volume decreases. The pressure (y -axis) and volume (x -axis) are plotted on a linearly scaled graph.

Which statement describes the plotted line?

- A** a curved line of increasing positive gradient starting at the origin of the graph
- B** a curved line with negative gradient of decreasing magnitude
- C** a straight line parallel to the pressure axis
- D** a straight line parallel to the volume axis
- E** a straight line of positive gradient starting at the origin of the graph

- 57** What is the highest common factor of 360, 500 and 700, given as the product of powers of its prime factors?
- A** 2×5
- B** $2^3 \times 5^3$
- C** $2^3 \times 3^2 \times 5^3 \times 7$
- D** $2^2 \times 5$
- E** $2^2 \times 3^2 \times 5 \times 7$
-
- 58** What is the sum of the solutions to the equation $\frac{3}{x} + \frac{2}{x-2} = 1$, where x is a real number and $x \neq 0$ and $x \neq 2$?
- A** -3
- B** -7
- C** 2.4
- D** 5
- E** 7
-
- 59** Which of the following expressions is equal to $\frac{8^{2n} \times 4^n}{2^n}$ for all integers n ?
- A** 2^{14n}
- B** 2^{6n+1}
- C** 2^{7n}
- D** 2^{2n+5}
- E** 2^{6n}

60

Two charged particles P and Q are 0.10 m apart. The charge on P is 1.50×10^{-7} C and the charge on Q is 1.50×10^{-7} C. Particle P experiences an electrostatic force of magnitude F because it is near to the charge on particle Q.

The distance between the two particles is increased to 0.20 m. The charge on P is increased to 4.50×10^{-7} C and the charge on Q is increased to 6.00×10^{-7} C.

What is the magnitude of the force that particle P experiences now?

- A $\frac{F}{4}$
- B $12F$
- C $6F$
- D $\frac{3F}{4}$
- E $3F$

